Build CAS information architecture with customized strategy

By Andy Tang

1. The introduction on CAS

The client I provide services is an academic research center, named the center of alcohol studies (CAS). It is a multidisciplinary institute dedicated to deep research, acquisition and dissemination of knowledge on psychoactive substance use and related phenomena with primary emphasis on alcohol use and consequences. Its researchers are professionals, who come from biological science, psychology, and sociology area, most of them being Ph.D.

1.1. The scheme

In order to build a competent website, a good work plan is definitely requested and related working process requires moving along step by step. The first, a detailed analysis on business strategy of client is required (Moville & Rosenfeld, 2006), and the context relating with it will be analyzed. The second, the problem of existing or current CAS website is demonstrated through deep analysis and comparison. The third, the gap between the business strategy and the target reached by CAS website, is exposed. The fourth, by comparing current website with some competitive ones and by doing a deep analysis, some improving advices are given in many important aspects on information architecture. The fifth, during the website building period, some usability test are made time by time, to ensure the website quality and related target to be realized.

1.2. The business strategy of CAS

The survival and development of CAS largely depends on the funds or grants coming from government or some contributors on research area. Hence, it is important for these contributors to make an effective measure on how CAS research results or performance can better push the research and application in this area forward. Without doubt, it is a benchmark on how much of funds should be delivered or provided to promote the research in progress. According to this, the key business or organization strategy of CAS is 3 points. The first, it is definitely necessary to generate professional research fruits as much as possible by use of all of the CAS resources. The second, it is very important to provide a wonderful learning and practice opportunities for whatever graduates, Ph.D., and Post Doctor, target on cultivating more researchers for this area. The third, it is important to disseminate or transfer some results of the research to the public, giving them a commonplace knowledge on the aspects of alcoholism.

1.3. The role of website on CAS

In order to carry out these strategies, it is not enough to merely hold powerful research strength. A flexible information exchange way is also a strong tool to promote the research in process. Without doubt, in information era, the

website is such kinds of a powerful platform to communicate with the similar research institutes and related contributors, simultaneously pushing the research process in advance. Without doubt, under the competitive research environments, a good website can act as good communication tools to demonstrate the research advantage and study ability. Furthermore, it can act as a promo, not only educate the public, but also promulgate the good fame of research institute efficiently.

1.4. The gap between the CAS business strategy and the role of its website.

However, the existing website of CAS, with a slow update speed and bad architecture, does not take effect as it should be. There are many deficiencies in current CAS website. The first, the existing website has no compact correlation with the business strategy the CAS holds. The second, the information organizational way is in disorder, not clear enough to attract the user interests. The third, the website can not provide enough information to support professional researcher to deepen their research. The fourth, the webpage are deficient of scannability, and it frightens the common user with a pile of nomenclature. The fifth, there are not enough resources in website to support students learning in this area. Some reference information is hidden deeply inside the website, hard to find. The sixth, the website use very boring layout on the introduction of professional researchers in CAS, transferring an inanimate impression to user.

2. Bridge the gap

It is very clear that there is a large gap between the CAS strategy and its existing website functions. Hence, an information architecture strategy, based on CAS business strategy, definitely requires being setup to bridge this gap. Finally, three key strategy points on information architecture are confirmed. The first, the website must provide professional services for professional researchers to strongly support their information communication and academic material sharing. The second, the website must provide colorful teaching resources for whatever graduates or Ph.D. to attract them into our study program. The third, the website should have the ability to gradually provide the knowledge to the public on how to accurately use alcohol and drugs in daily life and avoid trouble.

On the execution level of information architecture, there are some key components on architecture strategy required consideration, which separately are information architecture administration, technology integration, process emphasis, organizational or label system, document type identification, metadata field definition, and navigation system design (Moville & Rosenfeld, 2006). The is one extra point required attention that, on the aspect of aesthetics, the website must be succinct and attractive enough to pull user into the website and help to realize or reach the targets talked above.

2.1. Information architecture administration

On the aspect of architecture strategy, about 50 percent of resources and strengths should be put into the resources building which can be efficiently used by professional researchers. Because the CAS has many subordinate research

departments, it is definitely important to be familiar with their tasks, importance, and research requirements. Because the researchers on CAS have heavy research work loads, the interface design should not be made like a labyrinth. Because they are professionals, some special jargon or nomenclature can be easily understood by them, determining what kinds of Meta data or navigational system they request.

There are 30 to 35 percent of resources and strengths should be put into the resource building which provides services for students. On this aspect, the clearly structured teaching resources should be easily reached and obtained on the website by students, such like some courseware, reference book lists, journal articles, and some other things helpful on learning. The clear navigational system should be built for them to guide them to touch the related study information easily. The metadata should not be designed rather professional, instead should be designed as a bridge to promote students to understand the research concepts on alcohol studies.

There are 15 to 20 percent of resources and strengths should be put into the information collection, to promote public to have a basic understanding on how to deal with alcohol. The navigational system on this part must be very clearly designed and organized. The subject heading should be expressed by very easily understandable words.

2.2. Technology integration

In order to provide better services through website rebuilding, it is necessary to have a server with the ability to connect database into the front end. Search functions are definitely required by website. The content management task is allocated to different departments and professionals, for the reason of some very hardly understandable research contents mastered by them. The profiles of the researchers are provided on the basis of personalization to handle the different contexts.

2.3. Process emphasis

For the reason of already having strong top-down communication ability, the more strength is put into the bottom-up survey and investigation.

2.4. Label system

For three different user groups, who separately are professional researchers, students, and the public, the special label systems are designed to satisfy their individual information habits and provide understandable things in different layers.

2.5. Document type identification

For difference between user groups, there is different document emphasis to service them. For professional researchers, the document mainly consists of research journals and articles. For students, the document mainly consists of reference list and journal articles facing promoting academic ability, plus some pictures and PowerPoint files. For the public, the document mainly consists of easily scanning materials, including the format of audio, video, and common sense articles.

2.6. Metadata field definition

Different Metadata should be easy identified and face to different user groups, according to their knowledge level (Moville & Rosenfeld, 2006). Furthermore, there should have accurate and proper subject headings to classify different categories, and providing precise titles to give exact definition on documents.

2.7. Navigation system design

Facing different user groups, some personalized navigation systems are provided on different contexts. The top, left, and bottom navigational bar are provided on different focus to users to fit their browsing habits. Search functions may provide users a free option on searching subject headings or full texts. A pop up menu was added into navigational system in obvious position, on the one hand to save the space, on the other hand to give user a shortcut to important node page.

3. The post-design phase: research and comparison.

3.1. The use of usability test

In order to improve the usability of the website, the test on usability is made with the process of web building (Krug, 2006). Obviously, without the correction or rectification from the users, it is impossible to have a successful website. Hence, a detailed test on usability is held among the user in CAS. Three key people in the use of web are chosen from CAS to be representation of whole staffs, which are separately an information director, a researcher, and a journal editor. The question mainly focus on different layers of the website, such as whether our website are in coincidence with the CAS strategy, whether the navigational bar are suitable for our website, whether the layout of webpage are comfortable, whether the subject heading is a good representation on relative contents, whether the color design are harmonious in entire website, whether the website keep a good consistency, and so on. The returned test results are very helpful for the rectification of the web organization and architecture.

3.2. Modify an website and make it more effective

According to test results, an effective site structure in CAS should have some characteristics as below. The first, it should have a clear business strategy as a guide on information architecture (Krug, 2006). The second, its navigational bar should be clearly designed to inflect the business strategy in CAS. The third, the web layout should be clear enough and coincident with the use habit of service targets (Krug, 2006). The fourth, it should keep consistency to give user a relatively easy premonition on what kinds of content should be presented on which kinds of areas. The fifth, the subject heading and logo system should represent the related contents adequately. The sixth, the large paragraph contents should be divided and separately represented by clear subject heading or titles, forming a clear structure within pages.

3.3. The thinking on modification of website

In this course, there are so many invaluable things guiding me from a layman on web building into a specialist in information architecture. They include pre-phase user needs and behavior investigation and analysis, how to design organizational systems according to scheme, how to design a effective label system, how to setup a clear navigation system in coincidence with the strategy, how to pursue a timely usability test, how to let strategy become a main nexus on entire website building, how to use metadata to represent the contents in a better way, and so on.

3.3.1. Incorporate

For the contents or elements which do not exist in current CAS website, some new design, such as search system and a clear pop menu guides, are added to ensure holding good information architecture.

3.3.2. Modify

As for loose information organization on strategy of CAS, many web elements are modified to make them in coincidence with business strategy of CAS. The navigation system is redesigned to clearly provide service to different user group in CAS. The disordered web pages are reorganized and regrouped to better service them according to strategy. The subject headings and related contents are reorganized and relocated into suitable position in website. The metadata are redesigned to be a better representation on related aspects.

3.3.3. Improve

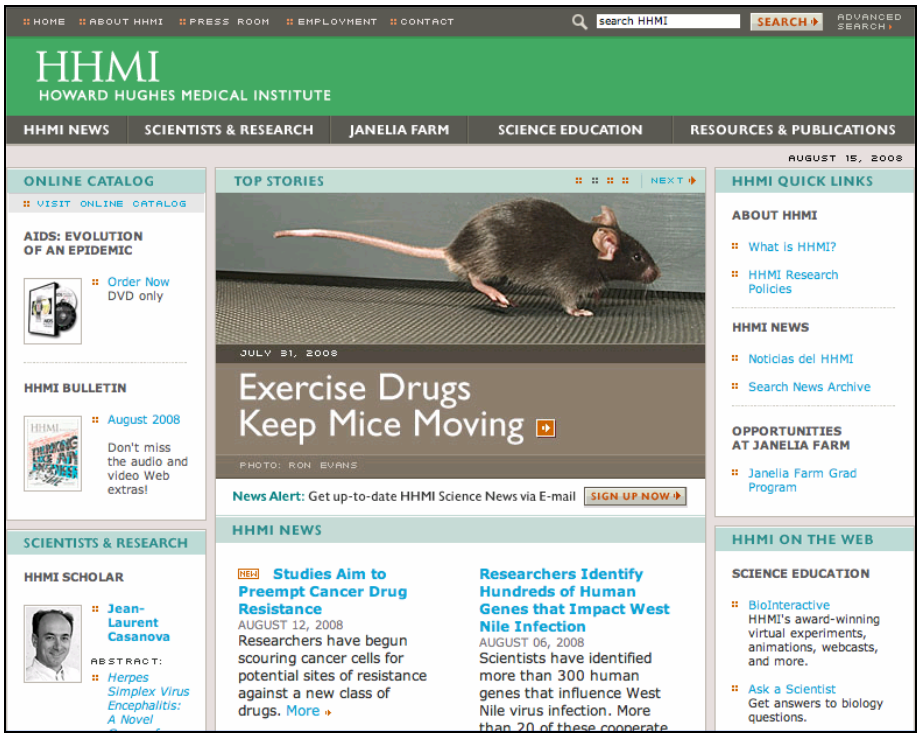
For the elements or components basically satisfying the existing website, some better idea and design are figured out to apply to related aspects and improve the performance of entire website. For example, some large paragraphs depictions are changed into succinct ones which provide different level of service to different group of users. Some pages on professional profiles are redesigned to give a wholly new and sharp feeling for the web visitors.

3.4. Learn from exemplar website

There are many professional websites which have perfect design and information organizational way with them. As in the same or neighborhood area, many websites provide a good exemplar on the directions of whatever modification or improvement of CAS website. The most perfect websites we used as benchmark or reference are www.hhmi.org, www.niaaa.nih.gov, www.nida.nih.gov, www.nlm.nih.gov, and so on.

There are many perfect elements which CAS website can borrow.

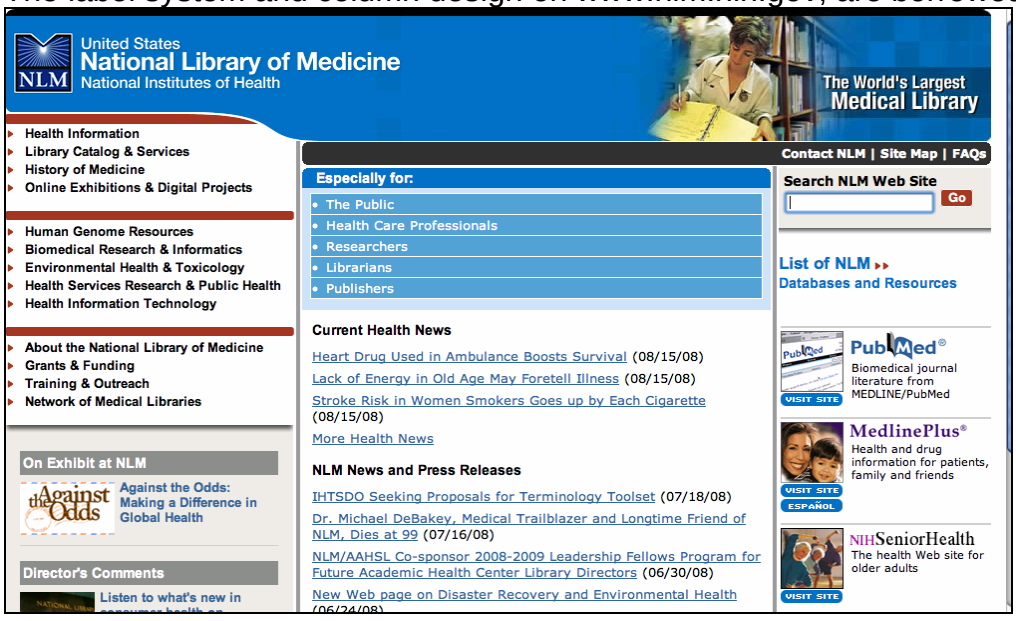
Through www.hhmi.org, a navigation system facing different user groups, based on business strategy, are adopted and borrowed by CAS. The Hughes Website definitely employ a good navigational system to separate its user groups clearly which separately are “scientists & research”, “Janelia Farm”, and “science education”, (See illustration below)



The way in organizing the subject heading and related contents on www.nida.nih.gov, are borrowed by CAS website.

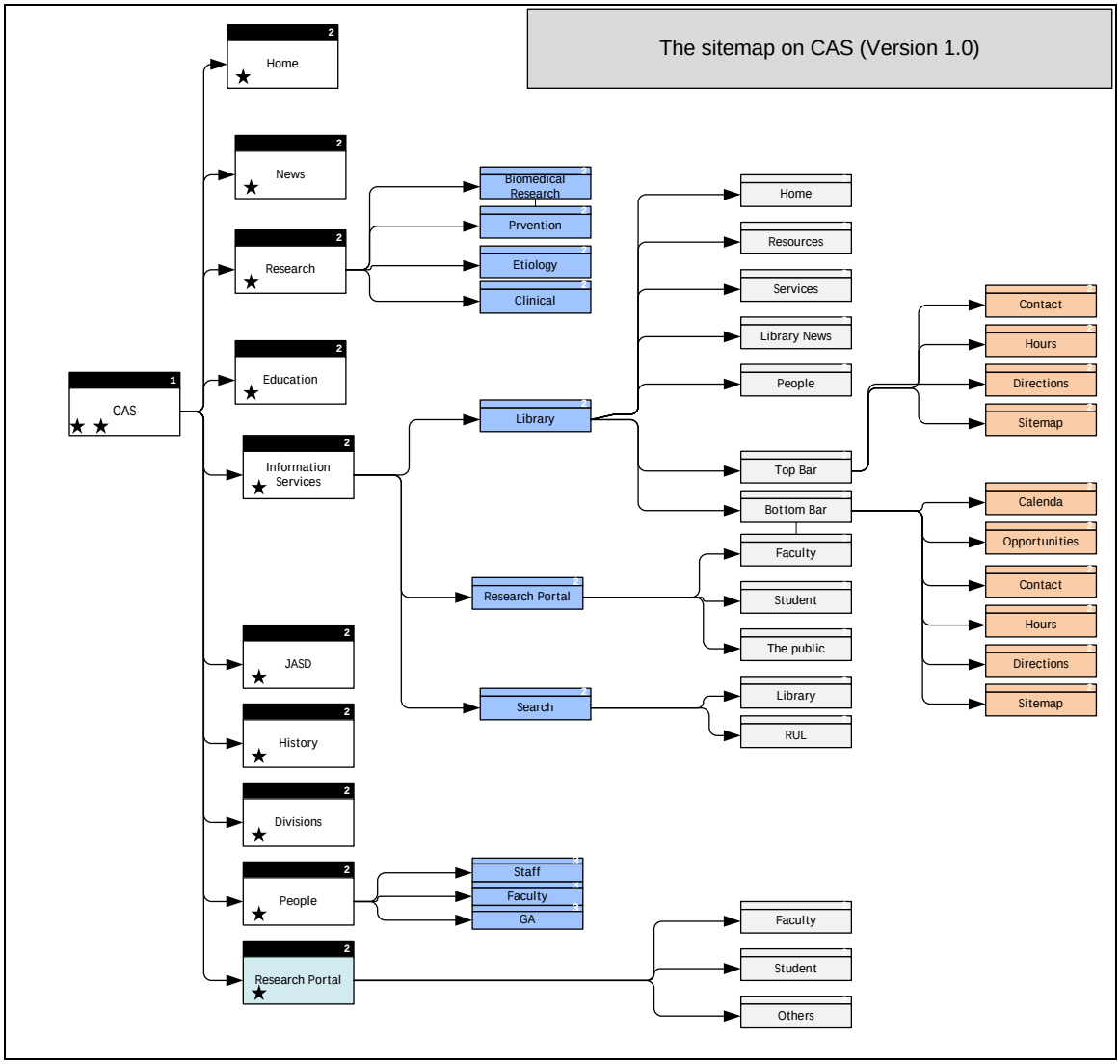


The label system and column design on www.nlm.nih.gov, are borrowed by CAS.



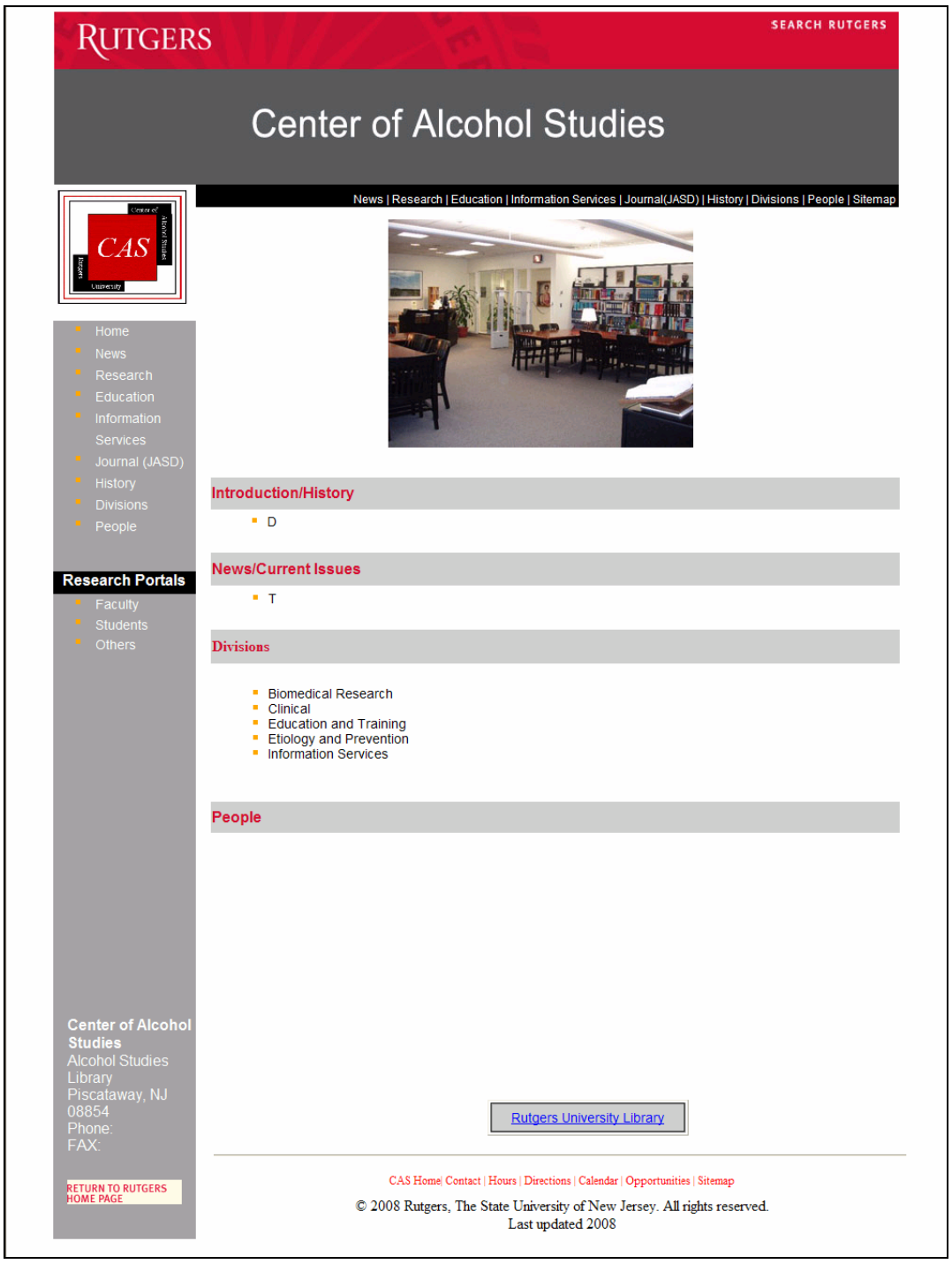
4. Illustrative diagrams

4.1. The main blue print



4.2. The wireframes on several main pages

4.2.1. The CAS page



4.2.2. The Information Services page

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Center of Alcohol Studies
Information Services

CAS Home | Contact | Hours | Directions | Sitemap

Library
- Library Home
- Resources
- Services
- Library News
- People
- Journal

Research Portals
- Faculty
- Students
- Others

Search
- CAS Library
- RUL

The Information Services Division houses the Center's extensive research library and the collections of the New Jersey Alcohol/Drug Resource Center and Clearinghouse. The Library's purpose is to document the world's scholarly, scientific, and professional English-language literature on psychoactive substance use, with a particular focus on beverage alcohol. A major task of the Library is to organize and disseminate information for scientists, scholars, and clinicians in the field.



News
- Dynamic News
- Newsletters

Library Services
- The Library's purpose is to document the world's scholarly, scientific, and professional English-language literature on psychoactive substance use, with a particular focus on beverage alcohol. A major task of the Library is to organize and disseminate information for scientists, scholars, and clinicians in the field. The Clearinghouse provides materials on alcohol/drug issues for educators, community workers, and other interested persons and groups in New Jersey.

Resources
- Resources mainly provide the information Coming from database, Federal Agencies online, Federal Clearinghouse online, Historical Resources, and other ones from Internet.

People

Journal of Studies on Alcohol & Drugs
- The Journal of Studies on Alcohol and Drugs, founded in 1940, publishes peer-reviewed manuscripts dealing with diverse aspects of alcohol and other substances of abuse. The range of materials includes biological, medical, epidemiological, social, psychological, and other aspects of alcohol and other drug use, abuse, and dependence.

Rutgers University Library

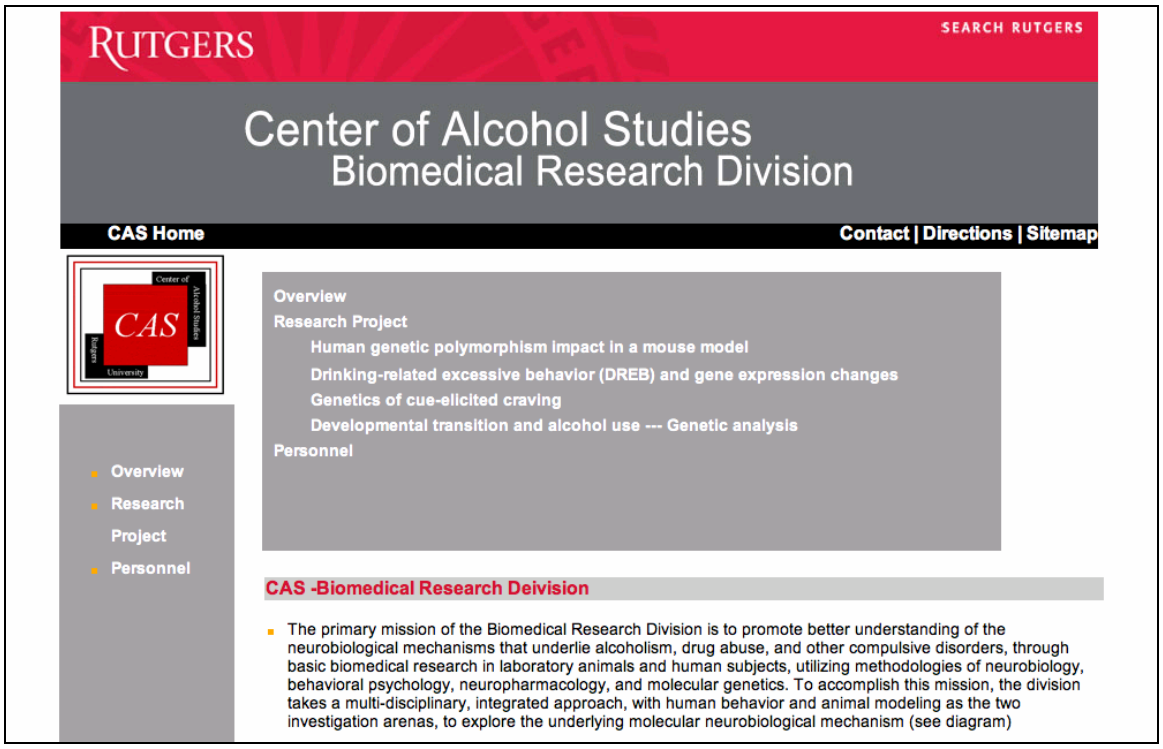
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Alcohol Studies Library
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4.2.3. Biomedical research division



4.2.4. **The profile of one of professional researchers**

RUTGERS SEARCH RUTGERS

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Dr. Johnson has been with the CAS since 1978. Her degree is in Sociology, specializing in substance use and mental health. She is a co-investigator of the Health and Human Development Project, a longitudinal study investigating substance use trajectories and risk and protective moderators and mediators among a community sample of adolescents and young adults. She is currently a principal investigator in the NIDA funded Rutgers Transdisciplinary Prevention Research Center, examining the efficacy of peer led and adult mentoring prevention programming within high schools.

Current Research

1 The evaluation of prevention/intervention programs.
2 The study of children from alcoholic and dysfunctional families.
3 The documentation of long-term outcomes of diverse drug using patterns.

Education

Ph.D	Rutgers University	Sociology	1985
M.A	American University	Policy Analysis	1977
B.A.	Douglass College	Political Science	1975

Representative Publications

Johnson, V. L. (in press). Fidelity of prevention programs. in N. Roget & G. Fisher (Eds.) Encyclopedia of substance abuse prevention, treatment, and recovery. Sage Publications.

Johnson, V.L, Holt, L.J. & Bry, B.H. (2008) Effects of an integrated preventative intervention on urban youth transiting into high school. Journal of Applied School Psychology, 24(2).

Valencia, E. & Johnson, V. (2008). Acculturation among Latino youth and risk for substance use: Issues of definition and measurement. Journal of Drug Issues, 38 (1), 37-68.

Warner, L.A., White, H.R. & Johnson, V.L. (2007). Alcohol initiation experiences and family history of alcoholism as predictors of problem drinking trajectories. Journal of Studies on Alcohol, 68(1), 56-65.

Johnson, V.L., Pandina, R.J. & Bry, B.H. Changes in social norms, harm perceptions and reasons for alcohol use among high school students participating in a peer led high school program. Presented at the Research Society on Alcoholism annual meeting. June 2008.

Johnson, V.L., Pandina, R.J., & Bry, B.H. A peer-led prevention program delivered to a predominately Hispanic high school. Presented at the Society for Prevention Research annual meeting. May 2008.

Johnson, V. L., Mun, E.Y. & Pandina, R.J. Results from a peer-led high school prevention program delivered in two inner- city schools. Paper presented at the Society for Research on Adolescence annual meeting. March 2008.

Johnson, V.L. Cultural adaptation, substance use harm perception and intent to use drugs among Hispanic high school students. Paper presented at the American Public Health Association annual meeting. November 2007.

4.3. Conclusion

It is definitely important to clarify the business strategy of an organization and the current status of the existing website. Without doubt, the next key step is to find a gap between them. Finally, it is necessary to bridge the gap by using effective ways on information architecture. Undoubtedly, by this way, a final product of website can satisfy the user requirements.

References

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